ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 2×10^{19}	n = 3.2 / 1.6×10^{-19}.
2	**(b)** 2 ohm	R / n.
3	**(d)** 1100 W	P = V I.
4	**(a)** 2 A	I = 600 / 300.
5	**(b)** 8 ohm	400 / 50.
6	**(c)** 484 ohm	V^2 / P.
7	**(b)** 36 J	W = V Q.
8	**(c)** 1 ohm	1/2 + 1/3 + 1/6 = 1.
9	**(c)** 1 kWh	2 x 0.5.
10	**(a)** 10 V	V = I R.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	n = I / e = 0.8 / 1.6×10^{-19} (1); n = 5×10^{18} (1).	1+1
12	1/R = 1/20 + 1/5 = 5/20 (1); R = 4 ohm (1).	1+1
13	Voltage gets divided among the appliances (1); If one fails the others also stop (1).	1+1
14	Rs = 25 ohm (1); I = 5 / 25 = 0.2 A (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	R = 3 + 4 + 2 = 9 ohm (1); I = 9 / 9 = 1 A (1); V = I R = 4 V (1).	1+1+1
16	R = rho L / A (1); Greater area gives lower resistance (1); Less heat I^2 R t, so the wire does not overheat (1).	1+1+1
17	Required equivalent R = 220 / 5 = 44 ohm (1); n resistors in parallel give 220 / n (1); n = 5 (1).	1+1+1
	OR: 1/R = 1/4 + 1/6 + 1/12 = 6/12 (1); R = 2 ohm (1); I = 6 / 2 = 3 A (1).	1+1+1
18	R = V^2 / P (1); 60 W bulb: R = 806.7 ohm (1); 100 W bulb: 484 ohm, so 60 W has more resistance (1).	1+1+1
	OR: Daily = 3 x 0.06 x 4 = 0.72 kWh (1); In 30 days = 21.6 kWh (1); 21.6 units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Parallel = 2 ohm; total = 6 ohm (1); (b) I = 12 / 6 = 2 A (1); (c) V = 2 x 2 = 4 V (1); (d) I in 6 ohm = 4/6 = 0.67 A (1); I in 3 ohm = 4/3 = 1.33 A (1).	1+1+1+1+1
	OR: H = I^2 R t (1); H = 4 x 50 x 600 (1); H = 1.2×10^5 J (1); In calories: 1.2×10^5 / 4.2 (1); = 28571 cal (1).	1+1+1+1+1
20	(a) R = 60 ohm (1); (b) I = 12 / 60 = 0.2 A (1); (c) P1 = I^2 R = 0.4 W (1); P2 = 0.8 W (1); P3 = 1.2 W (1).	1+1+1+1+1
	OR: (a) All four in series: 8 ohm (1); (b) All four in parallel: 2/4 = 0.5 ohm (1); (c) Two series pairs (4 ohm each) joined in parallel: 2 ohm (1); (d) Two in parallel (1 ohm) in series with one more 2 ohm: 3 ohm (1); Series raises and parallel lowers the resistance (1).	1+1+1+1+1

Mistakes to avoid:

- Adding the reciprocals incorrectly: for resistors in parallel $\frac{1}{R_p} = \sum \frac{1}{R}$ and the final answer is R_p, not the sum of reciprocals.
- Connecting an ammeter in parallel or a voltmeter in series; the ammeter goes in series and the voltmeter in parallel.
- Thinking that resistance is directly proportional to cross-sectional area; $R \propto \frac{1}{A}$ and $R \propto l$.
- Using $H = I^2Rt$ with t in minutes or hours; time must be in seconds (and energy in kWh needs power in kW and time in hours).
- Writing that in a series circuit the voltage across each resistor is the same; the **current** is the same in series and the **voltage** is the same in parallel.
- Forgetting that 1 kWh equals 3.6×10^6 J; and confusing the unit of resistivity (Ω m) with that of resistance (Ω).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 1 A	I = 16 / 16.
2	**(c)** 360 J	9 x 4 x 10.
3	**(a)** twice	Ohm's law.
4	**(c)** 7 ohm	5 + 2.
5	**(c)** four times	H is proportional to I^2.
6	**(a)** resistance	V / I = R.
7	**(a)** 2 V	I = 6/9; V = 2 A/3 x 3.
8	**(b)** 1100 W	P = V I.
9	**(d)** 90 ohm	When the length is tripled the area becomes one third: R x 9.
10	**(b)** dimmer	Voltage is shared.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	P = V I (1); P = 220 x 4 = 880 W (1).	1+1
12	Energy = 0.06 x 5 x 30 = 9 kWh (1); Cost = 9 x 6 = Rs 54 (1).	1+1
13	R = rho L / A = 1.7×10^{-8} x 100 / 10^{-6} (1); R = 1.7 ohm (1).	1+1
14	H = I^2 R t (1); R = 400 / (4 x 10) = 10 ohm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Each part = 4 ohm (1); 1/R = 4 x 1/4 = 1 (1); R = 1 ohm (1).	1+1+1
16	R is proportional to rho (1); Ratio = 1.1×10^{-6} / 1.6×10^{-8} (1); About 69 times larger for nichrome (1).	1+1+1
17	It must melt before the appliance wire gets hot (1); Excess current heats and melts it (1); Circuit breaks and protects the appliance (1).	1+1+1
	OR: I = 6 / 15 = 0.4 A (1); V1 = 2 V, V2 = 4 V (1); P = V I = 6 x 0.4 = 2.4 W (1).	1+1+1
18	Parallel part = 18 / 9 = 2 ohm (1); Total R = 4 ohm (1); I = 6 / 4 = 1.5 A (1).	1+1+1
	OR: I = 100 / 220 = 0.4545 A (1); t = 600 s (1); Q = I t = 272.7 C (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	R = rho L / A = 1.6×10^{-8} x 2 / 0.4×10^{-6} (1); R = 0.08 ohm (1); I = 4 / 0.08 = 50 A (1); P = V I = 200 W (1); Large current means heavy heating (1).	1+1+1+1+1
	OR: Current of the bulb = 3 / 6 = 0.5 A (1); Voltage across the resistor = 12 - 6 = 6 V (1); R = 6 / 0.5 (1); R = 12 ohm (1); Power dissipated in resistor = 3 W (1).	1+1+1+1+1
20	I = I1 + I2 and V is the same (1); 1/Rp = 1/R1 + 1/R2 (1); 1/Rp = 1/8 + 1/24 = 4/24 (1); Rp = 6 ohm (1); I = 12 / 6 = 2 A (1).	1+1+1+1+1
	OR: Volume is constant, so the area becomes half (1); Length is doubled (1); R = rho (2L) / (A/2) (1); New R = 4 x 5 (1); R = 20 ohm (1).	1+1+1+1+1

Mistakes to avoid:

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- Connecting an ammeter in parallel or a voltmeter in series; the ammeter goes in series and the voltmeter in parallel.
- Thinking that resistance is directly proportional to cross-sectional area; $R \propto \frac{1}{A}$ and $R \propto l$.
- Using $H = I^2Rt$ with t in minutes or hours; time must be in seconds (and energy in kWh needs power in kW and time in hours).
- Writing that in a series circuit the voltage across each resistor is the same; the **current** is the same in series and the **voltage** is the same in parallel.

- Forgetting that 1 kWh equals 3.6×10^6 J; and confusing the unit of resistivity (Ω m) with that of resistance (Ω).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Rs 36	6 kWh x 6.
2	**(c)** R/4	Area is four times.
3	**(d)** brighter	Each gets full voltage.
4	**(a)** 25 W	R = V^2 / P is highest for the least power.
5	**(c)** 4 ohm	R is proportional to L.
6	**(d)** 3 A	I = 12 / 4.
7	**(b)** 2.16×10^6 J	0.6 kWh x 3.6×10^6.
8	**(d)** its nature and temperature	Property of the material.
9	**(b)** 4 A	3 + 1.
10	**(a)** Electrical to heat	Joule heating.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	I = P / V (1); I = 55 / 220 = 0.25 A (1).	1+1
12	Area becomes one-fourth (1); Resistance becomes four times (1).	1+1
13	Same current flows through both (1); Coil has high resistance so more I^2 R heat (1).	1+1
14	I = 20 / 20 = 1 A (1); V = 1 x 15 = 15 V (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	H = I^2 R t (1); 9 x 5 x 120 (1); H = 5400 J (1).	1+1+1
16	R = 8 / 2 = 4 ohm (1); I = 20 / 4 (1); I = 5 A (1).	1+1+1
17	P = 1100 W (1); t = H / P (1); t = 300 s (1).	1+1+1
	OR: R = 220^2 / 40 = 1210 ohm (1); I = 40 / 220 = 0.18 A (1); At 110 V, P = 110^2 / 1210 = 10 W (1).	1+1+1
18	Slope gives resistance (1); R is inversely proportional to area (1); 2 ohm resistor is thicker (1).	1+1+1
	OR: V = I R = 24 V (1); P = V I (1); P = 48 W (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Geyser: 1.5 x 4 = 6 kWh (1); Fan: 0.1 x 8 = 0.8 kWh (1); Bulb: 0.06 x 6 = 0.36 kWh (1); Daily = 7.16 kWh; 30 days = 214.8 kWh (1); Bill = 214.8 x 7 = Rs 1503.6 (1).	1+1+1+1+1
	OR: R40 = 1210 ohm, R60 = 806.7 ohm (1); Series: same current (1); P = I^2 R so 40 W bulb glows brighter (1); Parallel: same voltage, P = V^2 / R (1); The 60 W bulb glows brighter (1).	1+1+1+1+1
20	R1 = 484 ohm; R2 = 1936 ohm (1); Total R = 2420 ohm (1); I = 220 / 2420 = 0.0909 A (1); Same current, P = I^2 R is larger for larger R (1); The 25 W lamp glows brighter (1).	1+1+1+1+1
	OR: (a) I = 1000 / 220 = 4.55 A (1); (b) R = 220^2 / 1000 = 48.4 ohm (1); (c) Daily energy = 1 x 2 = 2 kWh (1); Monthly = 60 kWh (1); In joules: 60 x 3.6×10^6 = 2.16×10^8 J (1).	1+1+1+1+1

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- Writing that in a series circuit the voltage across each resistor is the same; the **current** is the same in series and the **voltage** is the same in parallel.
- Forgetting that 1 kWh equals 3.6×10^6 J; and confusing the unit of resistivity (Ω m) with that of resistance (Ω).

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